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$$z^4 = 1$$

$$1 = \text{cis}(0)$$

$$\Rightarrow z_k = \text{cis}\left(\frac{0 + 2k\pi}{4}\right), \quad k = 0, 1, 2, 3$$

$$\Rightarrow z_0 = \text{cis}(0) = 1$$

$$\Rightarrow z_1 = \text{cis}\left(\frac{\pi}{2}\right) = i$$

$$\Rightarrow z_2 = \text{cis}(\pi) = -1$$

$$\Rightarrow z_3 = \text{cis}\left(\frac{3\pi}{2}\right) = -i$$

$$\Rightarrow \boxed{z_0 = 1, \quad z_1 = i, \quad z_2 = -1, \quad z_3 = -i}$$

## References